



LUNAR MODULE

This replica shows the layout of the interior of the lunar module in which Neil Armstrong and Buzz Aldrin descended to the Moon. They stood side by side, Armstrong on the left and Aldrin on the right, each with a restricted view through a triangular window.

special distinction that was supposed to mark them out from the other astronauts. Collins was there because a spinal problem had forced him to withdraw from the Apollo 8 mission, swapping with Jim Lovell who was originally slated for Apollo 11.

Armstrong, Aldrin, and Collins lifted off from launch pad 39A in Cape Kennedy (Canaveral) on July 16, 1969. During their three-day voyage into lunar orbit inside the command module Columbia, they had plenty of time to take in the astonishing visual spectacle. For Armstrong, the most dramatic moment was passing through the lunar shadow, with the Sun eclipsed and the Moon lit blue-gray by reflected light from the Earth. As they drew near

to the Moon, Collins was struck most vividly by its barrenness – he described it as a “monotonous rock pile” starkly contrasted to the “verdant valleys” of the planet that the astronauts had left behind.

The Moon landing took place on July 20, though from the point of view of Europe, Africa, and Asia, Armstrong did not walk on the Moon until July 21. The landing was not without incident. Armstrong and Aldrin climbed into the lunar module *Eagle* and, leaving Collins to his lonely vigil, separated from the command module, which was orbiting 60 miles (100km) above the Moon’s surface. They then dropped to an orbit of 8 miles (13km) altitude. The final powered descent from orbit to the surface of the Moon was the tensest phase of the entire mission. Each move was scrupulously monitored by ground controllers in Houston, ready to abort the landing at any moment if it



APOLLO 11 CREW

APOLLO 11 MISSION COMMANDER

Neil Armstrong (b. 1930) grew up in Wapakoneta, Ohio, and learned to fly at the age of 15.

He fought in the Korean War, flying off the carrier USS *Essex*, and went on to become one of America’s top test pilots at Edwards Air Force Base, pushing the X-15 to almost 4,000mph (6,500kph). Armstrong transferred to the NASA space program in 1962 from the USAF’s Dyna-Soar project for winged flight beyond the atmosphere. He had been into space once before, surviving the uncomfortable Gemini 8 mission in 1966.

Buzz Aldrin (b. 1930), the mission’s lunar landing module pilot, came from a military family, and his father had flown with the likes of Charles Lindbergh and Jimmy Doolittle. Like Armstrong, he was a Korean War veteran, but instead of going on to become a test pilot he took a doctorate at the Massachusetts Institute of Technology (MIT), for which he wrote a thesis on orbital rendezvous. Aldrin was regarded as an especially competitive man even by his colleagues – people not noted for self-effacement. He lobbied hard to be the first man to exit the landing module. Fellow Apollo 11 astronaut Michael Collins later wrote of Aldrin that he

“resents not being the first man on the Moon more than he appreciates being second.”

Mike Collins (b. 1930) was another scion of a military family – his father was an army general and his brother was a colonel. An Edwards AFB test pilot, he made a successful application to become a NASA astronaut in 1962. He seems to have accepted with equanimity his role as the sole crew member not to set foot on the Moon.



MOON TRAVELLERS

The Apollo 11 crew pose for an official photo, from left to right, mission commander Neil Armstrong, command module pilot Mike Collins, and lunar module pilot Buzz Aldrin.

“Here men from the planet Earth first set foot upon the Moon, July 1969 A.D. We came in peace for all mankind.”

INSCRIPTION FROM THE PLAQUE LEFT BY THE APOLLO 11 CREW ON THE MOON

MEN ON THE MOON

Edwin “Buzz” Aldrin walks on the Moon in July 1969. Reflected in his visor – which was goldplated to protect the astronaut’s eyes against the Sun – you can see the photographer, mission commander Neil Armstrong, and one leg of the Eagle lunar module. Beyond stretches the dusty plain of the Sea of Tranquility, marked for the first time by human footprints.